

17. Pandas program to split the DataFrame by school code and get mean, minimum, and maximum age for each school

Aim

To group the DataFrame by school code and find the mean, minimum, and maximum age of students.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing school code and age.
3. Group the data using school code.
4. Find the mean, minimum, and maximum age.
5. Display the result.

Program and output

```
import pandas as pd

# Create DataFrame
data = {
    "School_Code": ["S01", "S02", "S01", "S03", "S02", "S03"],
    "Student_Name": ["John", "Alice", "David", "Emma", "Robert", "Sara"],
    "Age": [15, 16, 14, 17, 15, 16]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Group by school code and calculate mean, min and max
result = df.groupby("School_Code")["Age"].agg(["mean", "min", "max"])

print("\nAge statistics for each school:")
print(result)
```

```

... Original DataFrame:
  School_Code Student_Name Age
0          S01         John  15
1          S02         Alice  16
2          S01         David  14
3          S03          Emma  17
4          S02        Robert  15
5          S03          Sara  16

Age statistics for each school:
      mean  min  max
School_Code
S01      14.5   14   15
S02      15.5   15   16
S03      16.5   16   17

```

Result

Thus, the mean, minimum, and maximum age for each school were calculated successfully.

18. Pandas program to split the DataFrame into groups based on school code and class

Aim

To divide the DataFrame into groups based on school code and class using Pandas.

Procedure

1. Import the Pandas library.
2. Create a DataFrame with school code and class.
3. Group the data using school code and class.
4. Display each group.
5. Verify the grouped data.

Program and output

```

import pandas as p

# Create DataFrame

data = {
    "School_Code": ["S01", "S01", "S02", "S02", "S01", "S03"],
    "Class": ["10A", "10B", "10A", "10B", "10A", "10A"],
    "Student_Name": ["John", "Alice", "David", "Emma", "Robert", "Sara"],
    "Age": [15, 16, 15, 16, 14, 17]
}

df = pd.DataFrame(data)

print("Original DataFrame:")

print(df)

```

```
# Group by School Code and Class

groups = df.groupby(["School_Code", "Class"])

print("\nGroups:")

for name, group in groups:

    print("\n", name)

    print(group)
```

```
Original DataFrame:
... School_Code Class Student_Name Age
0      S01    10A      John      15
1      S01    10B      Alice      16
2      S02    10A      David      15
3      S02    10B      Emma      16
4      S01    10A      Robert     14
5      S03    10A      Sara      17

Groups:

('S01', '10A')
  School_Code Class Student_Name Age
0      S01    10A      John      15
4      S01    10A      Robert     14

('S01', '10B')
  School_Code Class Student_Name Age
1      S01    10B      Alice      16

('S02', '10A')
  School_Code Class Student_Name Age
2      S02    10A      David      15

('S02', '10B')
  School_Code Class Student_Name Age
3      S02    10B      Emma      16

('S03', '10A')
  School_Code Class Student_Name Age
5      S03    10A      Sara      17
```

Result

Thus, the DataFrame was successfully divided into groups based on **school code and class**.